

The test results of the hydrophilicity and moisture absorption capacity of the material surface (see Supplementary Material Fig S1 for details) show that all formulations have a water contact angle of less than 90 ° (with hydrophilic characteristics, suitable for moist eye environments) and a water absorption rate of less than 14% (and the water absorption rate in PBS environment is slightly lower than that in deionized water), further verifying their basic adaptability as ophthalmic implant materials.

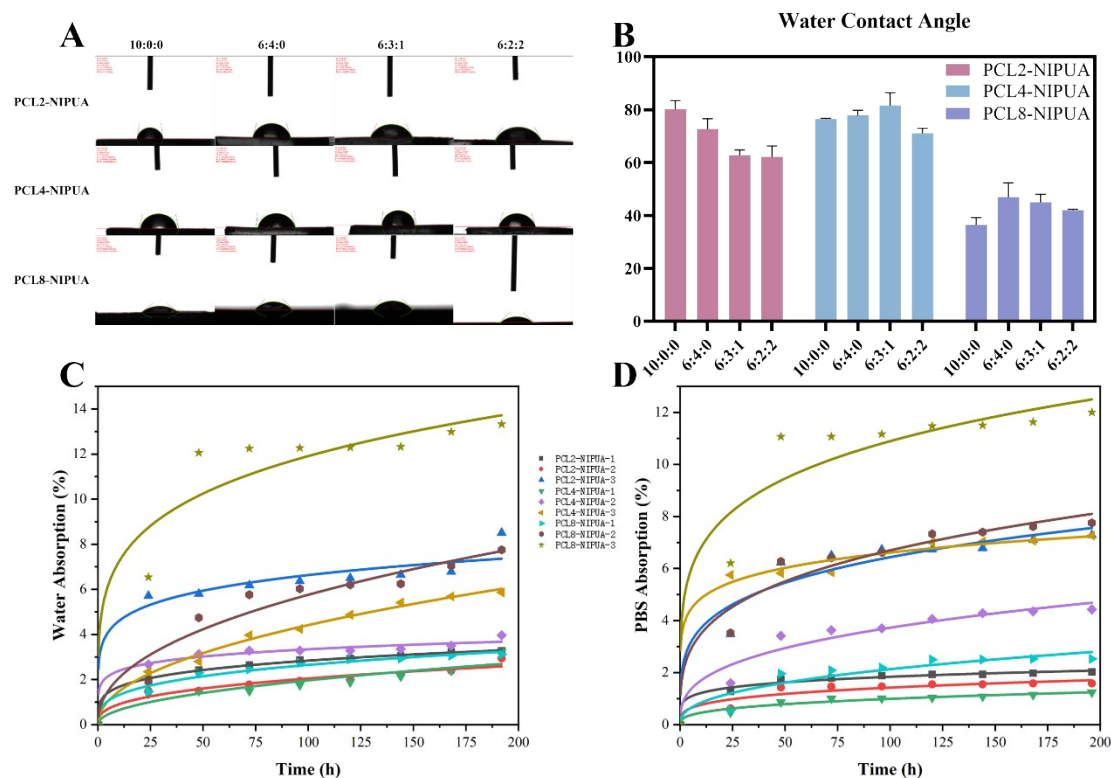


Figure S1 Physicochemical characterization of PCL-NIPUA resins. (A) Representative images of water droplet contact angle measurements. (B) Quantitative comparison of water contact angles across different resin formulations. (C) Time-dependent swelling behavior in deionized water. (D) Swelling kinetics in phosphate-buffered saline (PBS, pH 7.4).